```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

from sklearn.metrics import roc_curve, auc, precision_recall_curve
```

```
data = pd.read_csv('/content/data.csv')
data.drop(['id', 'Unnamed: 32'], axis=1, inplace=True)
```

```
# Encode labels (M = 1, B = 0)
data['diagnosis'] = data['diagnosis'].map({'M': 1, 'B': 0})

# Features and Target
X = data.drop('diagnosis', axis=1)
y = data['diagnosis']

# Standardize features
scaler = StandardScaler()
X = scaler.fit_transform(X)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = Sequential()
model.add(Dense(32, activation='relu', input_shape=(X.shape[1],)))
model.add(Dense(16, activation='relu'))
model.add(Dense(1, activation='sigmoid'))  # Binary classification
```

```
/usr/local/lib/python3.11/dist-packages/keras/src/layers/core/dense.py:87: UserWarning: Do not pass an `input_shape`/`input_dim` argumer
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
# 1. Train the model and store history

# Compile the model
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

history = model.fit(
    X_train, y_train,
    epochs=100,
    batch_size=16,
    verbose=1,
    validation_split=0.1
)
y_pred = (model.predict(X_test) > 0.5).astype(int)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))


# 2. Accuracy Plot
plt.figure(figsize=(10, 4))
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy Over Epochs')
plt.xlabel('Epoch')
```

```
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)
plt.show()

# 3. Loss Plot
plt.figure(figsize=(10, 4))
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Model Loss Over Epochs')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)
plt.show()

# 4. Confusion Matrix Heatmap
from sklearn.metrics import confusion_matrix
conf_mat = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6, 5))
sns.heatmap(conf_mat, annot=True, fmt="d", cmap='Blues', xticklabels=['Benign', 'Malignant'], yticklabels=['Benign', 'Malignant'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

# 5. ROC Curve
y_scores = model.predict(X_test).ravel()
fpr, tpr, thresholds = roc_curve(y_test, y_scores)
roc_auc = auc(fpr, tpr)
```

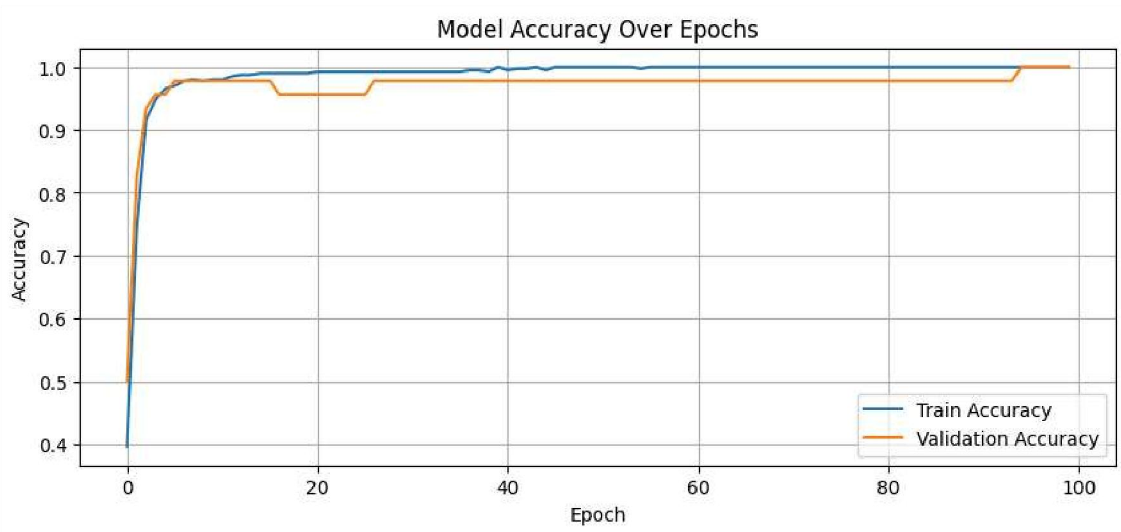
```
Epoch 1/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 2s 24ms/step - accuracy: 0.3990 - loss: 0.8462 - val_accuracy: 0.5000 - val_loss: 0.6397
Epoch 2/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.6375 - loss: 0.5727 - val_accuracy: 0.8261 - val_loss: 0.4680
Epoch 3/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 14ms/step - accuracy: 0.9298 - loss: 0.3908 - val_accuracy: 0.9348 - val_loss: 0.3258
Epoch 4/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 1s 23ms/step - accuracy: 0.9545 - loss: 0.2653 - val_accuracy: 0.9565 - val_loss: 0.2139
Epoch 5/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.9608 - loss: 0.1768 - val_accuracy: 0.9565 - val_loss: 0.1558
Epoch 6/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9720 - loss: 0.1472 - val_accuracy: 0.9783 - val_loss: 0.1283
Epoch 7/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9872 - loss: 0.1012 - val_accuracy: 0.9783 - val_loss: 0.1132
Epoch 8/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9759 - loss: 0.0959 - val_accuracy: 0.9783 - val_loss: 0.1046
Epoch 9/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9801 - loss: 0.0751 - val_accuracy: 0.9783 - val_loss: 0.0959
Epoch 10/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9896 - loss: 0.0602 - val_accuracy: 0.9783 - val_loss: 0.0921
Epoch 11/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9786 - loss: 0.0711 - val_accuracy: 0.9783 - val_loss: 0.0900
Epoch 12/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9860 - loss: 0.0571 - val_accuracy: 0.9783 - val_loss: 0.0864
Epoch 13/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9876 - loss: 0.0495 - val_accuracy: 0.9783 - val_loss: 0.0820
Epoch 14/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9923 - loss: 0.0457 - val_accuracy: 0.9783 - val_loss: 0.0779
Epoch 15/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9891 - loss: 0.0502 - val_accuracy: 0.9783 - val_loss: 0.0762
Epoch 16/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9926 - loss: 0.0434 - val_accuracy: 0.9783 - val_loss: 0.0753
Epoch 17/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9968 - loss: 0.0282 - val_accuracy: 0.9565 - val_loss: 0.0733
Epoch 18/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9801 - loss: 0.0576 - val_accuracy: 0.9565 - val_loss: 0.0747
Epoch 19/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9864 - loss: 0.0409 - val_accuracy: 0.9565 - val_loss: 0.0689
Epoch 20/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9904 - loss: 0.0352 - val_accuracy: 0.9565 - val_loss: 0.0702
Epoch 21/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9968 - loss: 0.0305 - val_accuracy: 0.9565 - val_loss: 0.0680
Epoch 22/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9883 - loss: 0.0383 - val_accuracy: 0.9565 - val_loss: 0.0682
Epoch 23/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9866 - loss: 0.0406 - val_accuracy: 0.9565 - val_loss: 0.0669
Epoch 24/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9918 - loss: 0.0338 - val_accuracy: 0.9565 - val_loss: 0.0667
Epoch 25/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9946 - loss: 0.0281 - val_accuracy: 0.9565 - val_loss: 0.0657
Epoch 26/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9951 - loss: 0.0209 - val_accuracy: 0.9565 - val_loss: 0.0656
Epoch 27/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9946 - loss: 0.0193 - val_accuracy: 0.9783 - val_loss: 0.0655
Epoch 28/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9936 - loss: 0.0265 - val_accuracy: 0.9783 - val_loss: 0.0664
Epoch 29/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9921 - loss: 0.0201 - val_accuracy: 0.9783 - val_loss: 0.0636
Epoch 30/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9981 - loss: 0.0179 - val_accuracy: 0.9783 - val_loss: 0.0650
Epoch 31/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9911 - loss: 0.0248 - val_accuracy: 0.9783 - val_loss: 0.0667
Epoch 32/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9943 - loss: 0.0149 - val_accuracy: 0.9783 - val_loss: 0.0664
Epoch 33/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9924 - loss: 0.0193 - val_accuracy: 0.9783 - val_loss: 0.0672
Epoch 34/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9915 - loss: 0.0170 - val_accuracy: 0.9783 - val_loss: 0.0653
Epoch 35/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9887 - loss: 0.0206 - val_accuracy: 0.9783 - val_loss: 0.0655
Epoch 36/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9940 - loss: 0.0159 - val_accuracy: 0.9783 - val_loss: 0.0652
Epoch 37/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9942 - loss: 0.0137 - val_accuracy: 0.9783 - val_loss: 0.0657
Epoch 38/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9951 - loss: 0.0141 - val_accuracy: 0.9783 - val_loss: 0.0656
Epoch 39/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9961 - loss: 0.0114 - val_accuracy: 0.9783 - val_loss: 0.0644
Epoch 40/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0100 - val_accuracy: 0.9783 - val_loss: 0.0627
Epoch 41/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9987 - loss: 0.0108 - val_accuracy: 0.9783 - val_loss: 0.0632
Epoch 42/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 0.9997 - loss: 0.0138 - val_accuracy: 0.9783 - val_loss: 0.0622
```

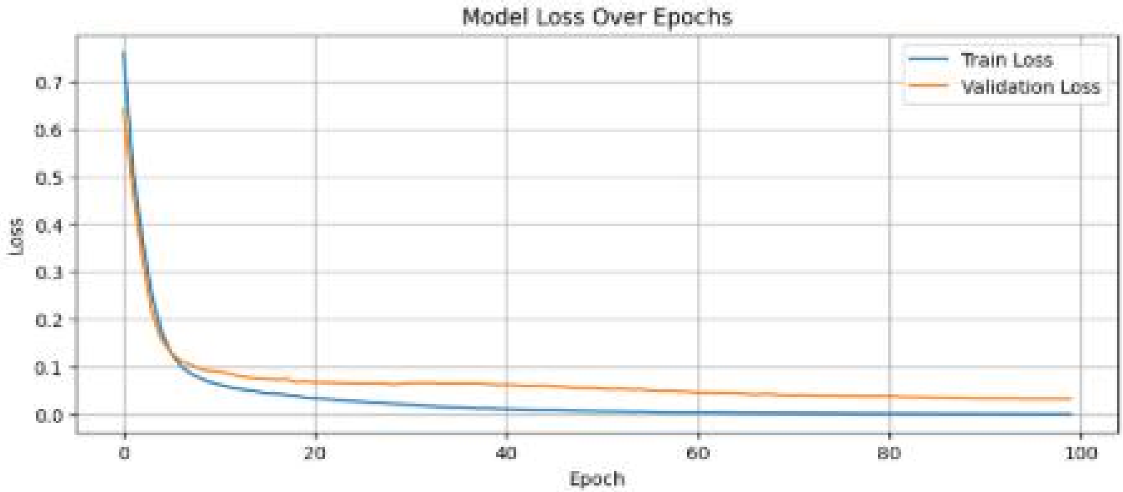
```
Epoch 43/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 0.9980 - loss: 0.0099 - val_accuracy: 0.9783 - val_loss: 0.0601
Epoch 44/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0082 - val_accuracy: 0.9783 - val_loss: 0.0605
Epoch 45/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9987 - loss: 0.0106 - val_accuracy: 0.9783 - val_loss: 0.0616
Epoch 46/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0122 - val_accuracy: 0.9783 - val_loss: 0.0590
Epoch 47/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0112 - val_accuracy: 0.9783 - val_loss: 0.0589
Epoch 48/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0054 - val_accuracy: 0.9783 - val_loss: 0.0567
Epoch 49/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0055 - val_accuracy: 0.9783 - val_loss: 0.0565
Epoch 50/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0077 - val_accuracy: 0.9783 - val_loss: 0.0564
Epoch 51/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0080 - val_accuracy: 0.9783 - val_loss: 0.0541
Epoch 52/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0075 - val_accuracy: 0.9783 - val_loss: 0.0535
Epoch 53/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0066 - val_accuracy: 0.9783 - val_loss: 0.0544
Epoch 54/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0038 - val_accuracy: 0.9783 - val_loss: 0.0528
Epoch 55/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9995 - loss: 0.0056 - val_accuracy: 0.9783 - val_loss: 0.0536
Epoch 56/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0052 - val_accuracy: 0.9783 - val_loss: 0.0510
Epoch 57/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0055 - val_accuracy: 0.9783 - val_loss: 0.0491
Epoch 58/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0049 - val_accuracy: 0.9783 - val_loss: 0.0507
Epoch 59/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0045 - val_accuracy: 0.9783 - val_loss: 0.0482
Epoch 60/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0065 - val_accuracy: 0.9783 - val_loss: 0.0481
Epoch 61/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0038 - val_accuracy: 0.9783 - val_loss: 0.0464
Epoch 62/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 1.0000 - loss: 0.0045 - val_accuracy: 0.9783 - val_loss: 0.0466
Epoch 63/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 1.0000 - loss: 0.0052 - val_accuracy: 0.9783 - val_loss: 0.0460
Epoch 64/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0043 - val_accuracy: 0.9783 - val_loss: 0.0461
Epoch 65/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 1.0000 - loss: 0.0020 - val_accuracy: 0.9783 - val_loss: 0.0449
Epoch 66/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 1.0000 - loss: 0.0039 - val_accuracy: 0.9783 - val_loss: 0.0434
Epoch 67/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 1.0000 - loss: 0.0029 - val_accuracy: 0.9783 - val_loss: 0.0420
Epoch 68/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 1.0000 - loss: 0.0039 - val_accuracy: 0.9783 - val_loss: 0.0434
Epoch 69/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0024 - val_accuracy: 0.9783 - val_loss: 0.0423
Epoch 70/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0023 - val_accuracy: 0.9783 - val_loss: 0.0414
Epoch 71/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0030 - val_accuracy: 0.9783 - val_loss: 0.0411
Epoch 72/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0028 - val_accuracy: 0.9783 - val_loss: 0.0414
Epoch 73/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0029 - val_accuracy: 0.9783 - val_loss: 0.0407
Epoch 74/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0024 - val_accuracy: 0.9783 - val_loss: 0.0403
Epoch 75/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0036 - val_accuracy: 0.9783 - val_loss: 0.0409
Epoch 76/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 7ms/step - accuracy: 1.0000 - loss: 0.0022 - val_accuracy: 0.9783 - val_loss: 0.0387
Epoch 77/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0013 - val_accuracy: 0.9783 - val_loss: 0.0389
Epoch 78/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0023 - val_accuracy: 0.9783 - val_loss: 0.0384
Epoch 79/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0023 - val_accuracy: 0.9783 - val_loss: 0.0381
Epoch 80/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0026 - val_accuracy: 0.9783 - val_loss: 0.0383
Epoch 81/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0021 - val_accuracy: 0.9783 - val_loss: 0.0385
Epoch 82/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0019 - val_accuracy: 0.9783 - val_loss: 0.0377
Epoch 83/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0017 - val_accuracy: 0.9783 - val_loss: 0.0366
Epoch 84/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0017 - val_accuracy: 0.9783 - val_loss: 0.0367
Epoch 85/100
```

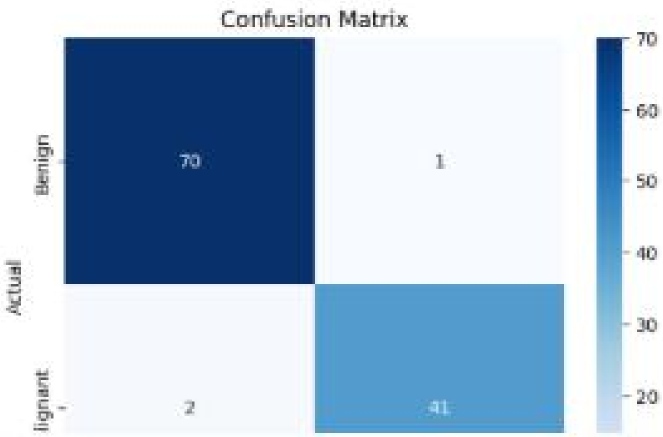
```
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0014 - val_accuracy: 0.9783 - val_loss: 0.0367
Epoch 86/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0020 - val_accuracy: 0.9783 - val_loss: 0.0357
Epoch 87/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0016 - val_accuracy: 0.9783 - val_loss: 0.0360
Epoch 88/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0012 - val_accuracy: 0.9783 - val_loss: 0.0352
Epoch 89/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0020 - val_accuracy: 0.9783 - val_loss: 0.0353
Epoch 90/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0024 - val_accuracy: 0.9783 - val_loss: 0.0338
Epoch 91/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0017 - val_accuracy: 0.9783 - val_loss: 0.0342
Epoch 92/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0015 - val_accuracy: 0.9783 - val_loss: 0.0339
Epoch 93/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 0.0013 - val_accuracy: 0.9783 - val_loss: 0.0343
Epoch 94/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 5ms/step - accuracy: 1.0000 - loss: 8.8895e-04 - val_accuracy: 0.9783 - val_loss: 0.0341
Epoch 95/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 0.0011 - val_accuracy: 1.0000 - val_loss: 0.0330
Epoch 96/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 9.4648e-04 - val_accuracy: 1.0000 - val_loss: 0.0333
Epoch 97/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step - accuracy: 1.0000 - loss: 8.6265e-04 - val_accuracy: 1.0000 - val_loss: 0.0333
Epoch 98/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 9.7917e-04 - val_accuracy: 1.0000 - val_loss: 0.0334
Epoch 99/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 9.4261e-04 - val_accuracy: 1.0000 - val_loss: 0.0330
Epoch 100/100
26/26 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 1.0000 - loss: 0.0010 - val_accuracy: 1.0000 - val_loss: 0.0331
4/4 ━━━━━━━━━━━━━━━━━━━━ 0s 17ms/step
Accuracy: 0.9736842105263158
[[70  1]
 [ 2 41]]
              precision    recall  f1-score   support

           0       0.97      0.99      0.98        71
           1       0.98      0.95      0.96        43

    accuracy                           0.97       114
   macro avg       0.97      0.97      0.97       114
weighted avg       0.97      0.97      0.97       114
```




Model Loss Over Epochs
Train Loss
Validation Loss
Loss
Epoch
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
20
40
60
80
100


Confusion Matrix
Benign
Actual
lignant
70
1
2
41
70
60
50
40
30
20